SHARDA UNIVERSITY
Beyond Boundaries

B.Tech Project Evaluation-2 ,VIIth Sem

Project Title - Intelligent DevOps Monitoring & Incident Response System

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Contents of the Presentation

- Title of the Project
- Introduction to the Project
- Motivation for the Project Problem Statement / Objectives of the Project
- Problem Domain
- Literature Survey / Existing System
- Team Members & Workload Distribution Meeting Details
- Project Planning / Timeline
- Uniqueness of the Idea
- Expected Outcome
- Conclusion

Introduction to the Project

Intelligent DevOps Monitoring System:
The Intelligent DevOps Monitoring & Incident Response System is an automated platform designed to monitor applications and infrastructure, identify abnormal behavior, analyze incidents and assist in resolving them.

The system combines:

- Real-Time Monitoring
  Continuous collection of infrastructure and application metrics.
- Anomaly Detection
  AI/ML models identify unusual system behavior.
- Intelligent Incident Analysis
  LLM analyzes monitoring data and identifies possible causes.
- Automated Response
  A response engine recommends or executes predefined recovery actions.
- Incident Visualization
  A dashboard displays system health, alerts and incidents.

Key Idea
Instead of only generating alerts, the system converts monitoring data into **intelligent diagnosis and actionable incident responses**.

Motivation for the Project

Why do we need this?
Modern applications generate large amounts of operational data through:

- Server Metrics
- Application Logs
- CPU & Memory Usage
- Network Metrics
- Container Metrics
- Service Health Data

However, this information often becomes:

- Difficult to analyze manually
- Too large to monitor continuously
- Difficult to correlate
- Responsible for alert fatigue
- Slow to convert into useful actions

Motivation:
We wanted to build a system where monitoring data can be continuously collected, intelligently analyzed and converted into actionable incident responses.

Our Goal:
Move from:
"An alert has occurred." to
"An issue has been detected, analyzed and an appropriate responsehas been identified."

Problem Statement / Objectives

Problem Statement:
Traditional monitoring systems primarily collect metrics and generate alerts, but they often require engineers to manually analyze incidents, identify root causes and execute corrective actions.

There is a need for a unified system that provides:
Monitoring + Anomaly Detection + AI Analysis + Incident Response
while maintaining reliability and operational control.

Objectives:
- **Enable real-time monitoring** of infrastructure and applications.
- **Detect system anomalies and incidents** automatically.
- **Analyze incidents and identify probable root causes** using AI.
- **Provide intelligent remediation recommendations** for detected incidents.
- **Improve incident response and system reliability** through automation and centralized monitoring.

Problem Domain

DevOps Monitoring + AIOps + Artificial Intelligence

The project lies at the intersection of:

- **DevOps Monitoring:-**Continuous monitoring of applications, services and infrastructure.
- **Anomaly Detection:-**Identifying unusual patterns in system metrics and behavior.
- **Artificial Intelligence:-**Using LLM-based reasoning to interpret incidents and generate useful responses.
- **Incident Management:-**Identifying, prioritizing and tracking infrastructure incidents.
- **Automation:-**Executing predefined actions to reduce manual intervention.
- **Distributed Systems:-**The platform integrates multiple components for monitoring, communication, AI analysis and response execution.

Literature Survey / Existing System

Existing System	Strength	Limitation
Prometheus	Powerful time-series monitoring and metric collection	Primarily focuses on monitoring; limited intelligent incident diagnosis and automated remediation
Grafana	Excellent visualization, dashboards and alert monitoring	Requires external systems for deeper incident analysis and automated response
ELK Stack	Strong centralized log collection and log analysis	Large-scale setup can be complex; limited autonomous incident resolution
Datadog	Integrated monitoring, logs, traces and alerting	Primarily relies on predefined monitoring and alerting workflows; limited customization for project-specific autonomous remediation
Our Platform	Monitoring + LLM-based diagnosis + incident detection + intelligent response/remediation	Requires development and testing of a reliable test bench and response mechanism

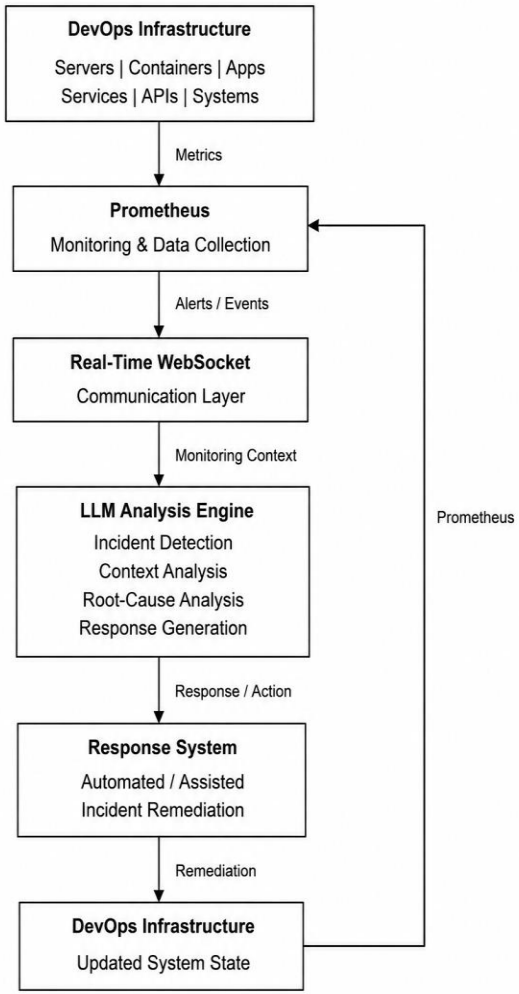
Research Gap

Most existing DevOps solutions primarily focus on:
Monitoring & Visualization or **Alerting & Incident Management or Observability**

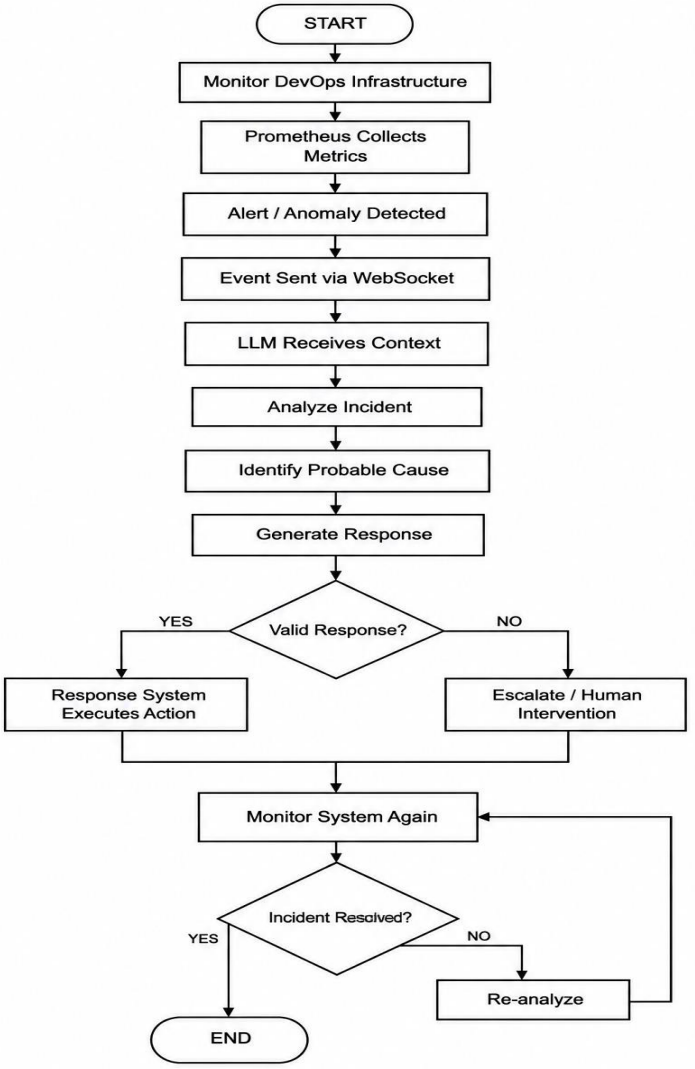
Our project aims to combine:
Real-time Monitoring + Incident Detection + LLM-based Analysis + Root Cause Identification + Response/Remediation

Project Architecture and Workflow

Architecture



Workflow



Team Members & Workload Distribution

Team Member	Major Responsibility
Muhammad Shayaan Ali	Backend / AI Integration/DevOps
Riddhima Sharma	Frontend / Dashboard
Shruti Tyagi	Monitoring / Testing

Overall Work Distribution

- **Frontend :** React / Next.js, Dashboard, UI/UX
- **Backend :** Python / FastAPI, REST APIs, Incident Processing
- **Monitoring :** Prometheus, Metrics Collection
- **Communication :** WebSockets
- **AI :** LLM, Prompt Engineering, Incident Analysis
- **Response :** Automated Remediation Engine
- **Infrastructure :** Docker, Containerized Services
- **Testing :** Unit, Integration and System Testing

Meeting Details

Project Patent

Weekly Mentorship Meetings with Mr. Ashish Kumar (Ex - Uni Mentor) in

Date	Key Discussion / Work Done
19 Apr 2026 (Week 1)	Discussed the initial **project idea, problem statement and objectives** for the Intelligent DevOps Monitoring & Incident Response System.
22 May 2026 (Week 2)	Discussed the **overall system architecture and workflow**. Planned the use of **Prometheus for monitoring and metric collection**, with the collected data being passed to an intelligent analysis component for identifying DevOps issues.
01 Aug 2026 (Week 3)	Reviewed the project approach and discussed the creation of a **controlled DevOps test bench/testbed**.

Project Progress

Discussions with Xebia Industry Mentor Mr. Raygun Jose , Online Meetings on GoogleMeet

Date	Key Discussion Points
1 Aug 2026 (Meeting 1)	Discussed the project from an **industry-oriented perspective**, particularly the need for a realistic and repeatable **DevOps test bench**.
17 Aug 2026 (Meeting 2)	Discussed recreating optional problems, collecting metrics from Prometheus, passing the monitoring data to an LLM for incidental analysis, and developing a response/remedation mechanism

Project Planning / Timeline

Phase 1 -Project Setup & Research
- Problem identification
- Literature survey
- Technology selection
- System requirements
- Initial architecture

Phase 2 -Monitoring Infrastructure
- Prometheus setup
- Metrics collection
- Monitoring endpoints
- Alert generation
- Initial dashboard

Phase 3 -AI & Incident Analysis
- Data processing pipeline
- WebSocket communication
- LLM integration
- Incident classification
- Root-cause analysis

Phase 4 -Automated Response
- Response engine
- Remediation workflows
- Action validation
- Incident resolution tracking

Phase 5 -Testing & Deployment
- Integration testing
- Performance testing
- Failure simulation
- Docker deployment
- Final documentation

Uniqueness of the Idea

What makes our system different?

The uniqueness lies in combining multiple DevOps capabilities into one intelligent workflow.

- **Intelligent Monitoring**
  The system continuously observes infrastructure rather than relying only on manual monitoring.

- **AI-Based Incident Analysis**
  LLM analyzes monitoring information and provides an understandable explanation of the incident.

- **Proactive Incident Understanding**
  System attempts to identify abnormal behavior before engineers manually investigate the issue.

- **Automated Response**
  Predefined remediation actions can be triggered for suitable incidents.

- **Unified DevOps Workflow**
  Monitoring, analysis and response are connected instead of operating as isolated tools.

Core Concept

Monitor → Detect → Analyze → Diagnose → Respond → Verify

Expected Outcome

At the end of the project, users should be able to:

- Monitor infrastructure in real time.
- View system health through a dashboard.
- Detect abnormal system behavior.
- Receive intelligent incident alerts.
- Analyze incidents using AI.
- Identify possible root causes.
- Generate incident reports.
- Receive recommended remediation actions.
- Execute predefined automated responses.
- Track incident history and resolution.

Expected Benefits

- Reduced manual monitoring effort.
- Faster incident identification.
- Faster troubleshooting.
- Reduced response time.
- Improved system reliability.
- Better visibility into infrastructure health.
- More efficient DevOps operations.

Patent filing in progress

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INVENTION DISCLOSURE FORM

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IMPORTANT

Please note that patent is always about a solution to an existing technical/industrial problem. As such, draft patent document (IDF) should be crafted in a manner so as to distinctly highlight the existing technical problem, the technical advancement proposed by the inventor(s), and the manner in which the proposed advancement solves the existing technical problem.

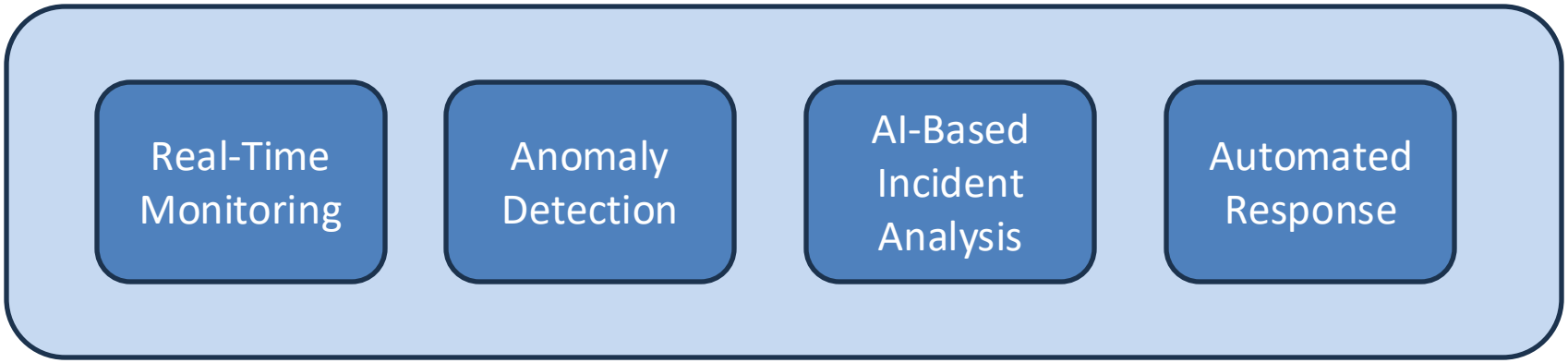
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Conclusion

The Intelligent DevOps Monitoring & Incident Response System aims to transform traditional monitoring into an intelligent and automated operational process.

By combining:



The platform provides a unified approach to identifying, understanding and responding to infrastructure incidents.

The architecture is designed to be
Intelligent Automated Real-Time Scalable DevOps-Oriented

Final Statement:
Our goal is not simply to build another monitoring dashboard, but to build an intelligent system that can **understand incidents and assist in resolving them with minimal manual intervention.**